Introduction to Machine Learning

Session 4 – Lecture Notes

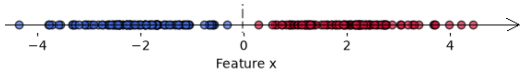
Logistic Regression

1D

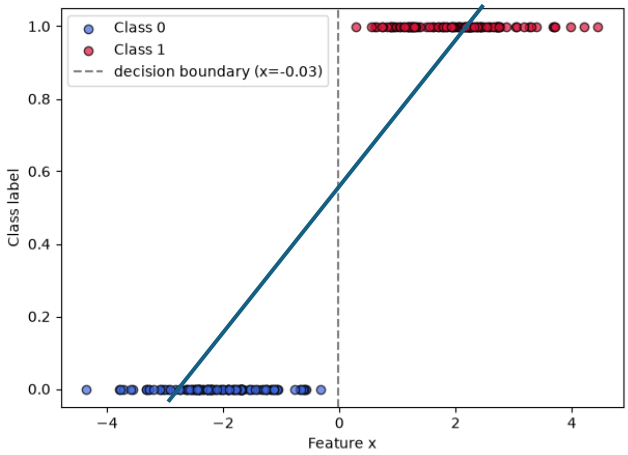
$$y = wx + b$$

x – numbers,

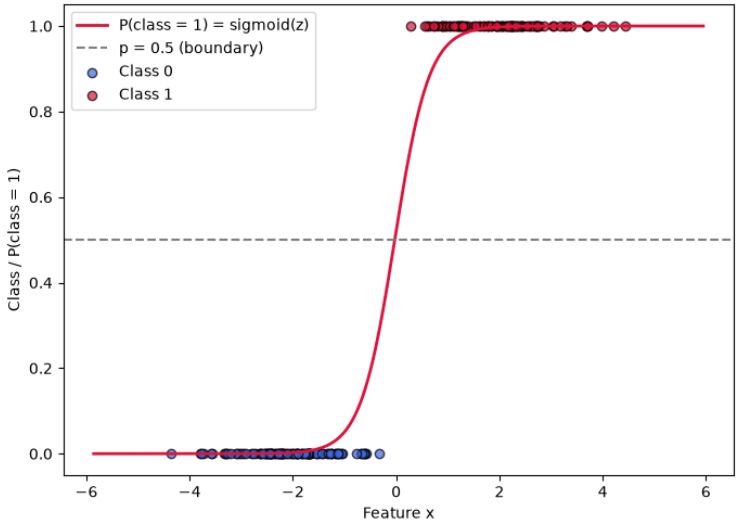
y – class/category



Expanding to 2D:



We are searching for 'maximum likelihood'. In Logistic regression we employ sigmoid function to map the relationship.



$$\hat{y} = \sigma(wx + b) = \frac{1}{1 + e^{-z}}, \quad z = wx + b$$

Loss - binary cross entropy

$$L = -[\, y \cdot log(\hat{y}) + (1 - y) \cdot log(1 - \hat{y}) \,] \text{ (one datapoint)}$$

$$L = -\frac{1}{n}\sum_{i=1}^{n}[\, y_i \cdot log(\hat{y}_i) + (1 - y_i) \cdot log(1 - \hat{y}_i) \,]$$

Gradient

$$\frac{\partial L}{\partial \hat{y}} = \frac{1}{n}\sum_{i=1}^{n}\frac{\hat{y}_i - y_i}{\hat{y}_i(1 - \hat{y}_i)}$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\hat{y}_i - y_i}{\hat{y}_i(1 - \hat{y}_i)} \cdot \hat{y}_i(1 - \hat{y}_i)\right] = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w} = \frac{1}{n}\sum_{i=1}^{n} x_i(\hat{y}_i - y_i)$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial b} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)$$

In matrix form: $\partial w = \frac{1}{n} \cdot X^T(\hat{y} - y) \quad \partial b = \frac{1}{n} \cdot \Sigma(\hat{y} - y)$

For more features (higher dimensions):

$$\frac{\partial L}{\partial w_1} = \frac{1}{n} \cdot \sum_{i=1}^{n} x_{1,i}(\hat{y}_i - y_i)$$

$$\vdots$$

$$\frac{\partial L}{\partial w_m} = \frac{1}{n} \cdot \sum_{i=1}^{n} x_{m,i}(\hat{y}_i - y_i)$$

Overfitting Trap

In theory we can establish number of parameters equal to the number of points in the dataset.

Now, if we employ the closed-form solution, we get to a point where our curve hits every point in the dataset. Which renders the model useless – data memorization not modelling.

Regularisation

1. Stop when validation goes up.
2. In training regularisation – keeping the coefficients modest

Ridge

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \; mean\; square\; error$$

$$L_{Ridge} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \cdot \sum_{j=1}^{m} w_j{}^2$$

$$\frac{\partial L}{\partial w} = -\frac{2}{n}X^{\mathrm{T}}(y - \hat{y}) + 2\lambda w = -\frac{2}{n}\sum_{i=1}^{n} x_i(y_{\mathrm{i}} - \hat{y}_{\mathrm{i}}) + 2\lambda \cdot w$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n}\sum_{i=1}^{n}(y_{\mathrm{i}} - \hat{y}_{\mathrm{i}})$$

λ = 0 → closer to the unconstrained / unpenalized solution

λ = ∞ → closer to a flat line

Vector notation:

$$w = (X^{\mathrm{T}}X + \lambda I)^{-1}X^{\mathrm{T}}y$$

(I = identity matrix)

Choosing λ - Log-spaced values that we test through validation:

1. Fit model with each λ separately
2. Score each on the validation data set
3. λ winner is the one with the best validation score
4. Last, score on the test set.

Lasso Regularisation

$$L_{Lasso} = \frac{1}{n}\sum_{i=1}^{n}(y_{\mathrm{i}} - \hat{y}_{\mathrm{i}})^2 + \lambda \cdot \sum_{j=1}^{m}|w_{\mathrm{j}}|$$

$$\frac{\partial L}{\partial w_j} = -\frac{2}{n}\sum_{i=1}^{n} x_i(y_{\mathrm{i}} - \hat{y}_{\mathrm{i}}) + \lambda sign(w_j)$$

No closed form for this regularisation, due to a derivative issue (|w| is not differentiable at 0).

Similar for logistic regression:

- Penalty L1 (lasso penalty)
- Penalty L2 (ridge penalty)